WEEK 1 · DAY 3 · OVERVIEW

The Price-Yield Relationship



Created by **Jeff Lenamon**

ABOUT 40 MIN WITH EXERCISES / 8 SECTIONS / 5 EXERCISES

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The Desk Lead's Note

Notebook 02 priced the four: Pellwick above 100, Dunmarrow on it, Quorvane and Halvering below. Tell me why each sits where it does. Show me how each price moves as yields move, and which bond moves most. If yields never move, where does each price go? A dealer offers an annual-pay bond at 5.45 against Quorvane's 5.40: which yields more? Put the restatement in bondmath, docstring first, prove it against Excel's EFFECT and NOMINAL, commit with a message that says why, and take it home.

ABOUT 40 MIN WITH EXERCISES / 8 SECTIONS / 5 EXERCISES

Your Route Through Day 3

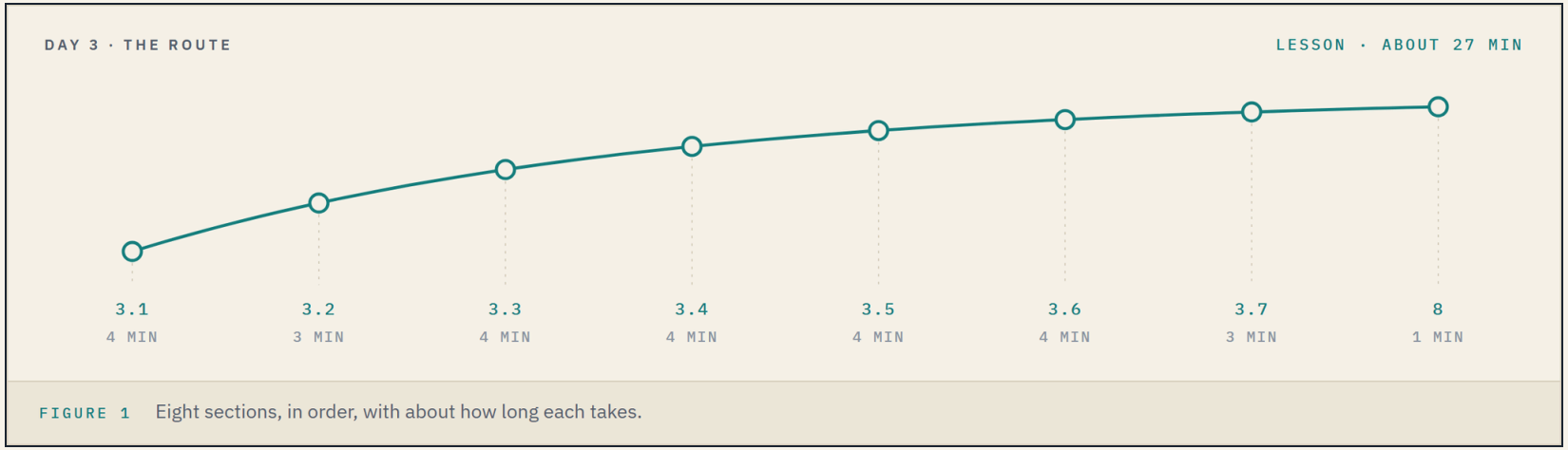


FIGURE 1 Eight sections, in order, with about how long each takes.

	SECTION	ABOUT
3.1	Par, Premium, and Discount	4 min
3.2	The Price-Yield Curve, Charted	3 min
3.3	Coupon and Maturity: Which Bond Moves More	4 min
3.4	Pull to Par on Coupon Dates	4 min

	SECTION	ABOUT
3.5	One Yield, Three Quotes	4 min
3.6	`convert_yield` in Your Module	4 min
3.7	A Commit Message That Says Why	3 min
8	Bring It Home	1 min

Time. About 27 minutes for the lesson, then about 15 for the five exercises.

3.1 Par, Premium, and Discount

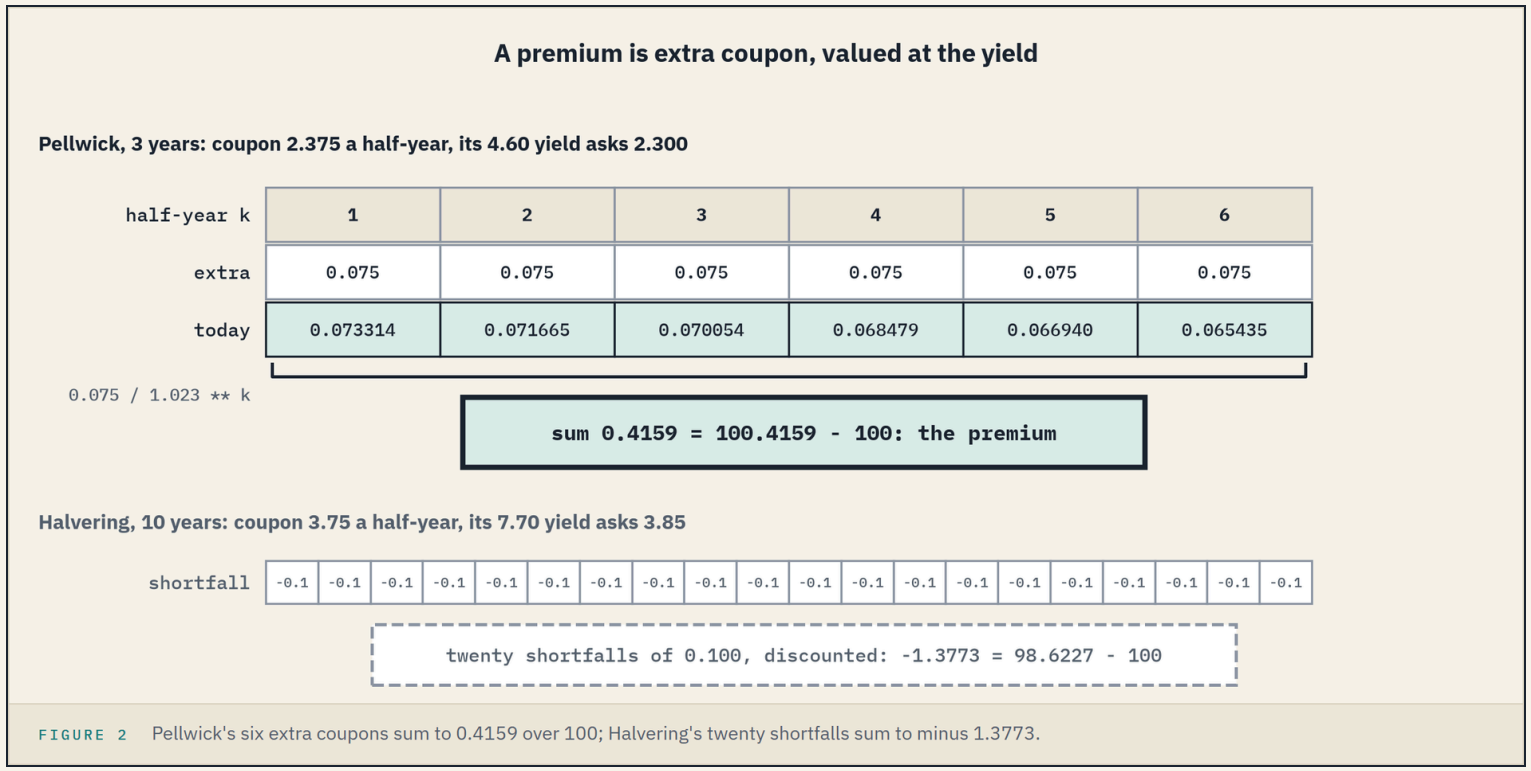


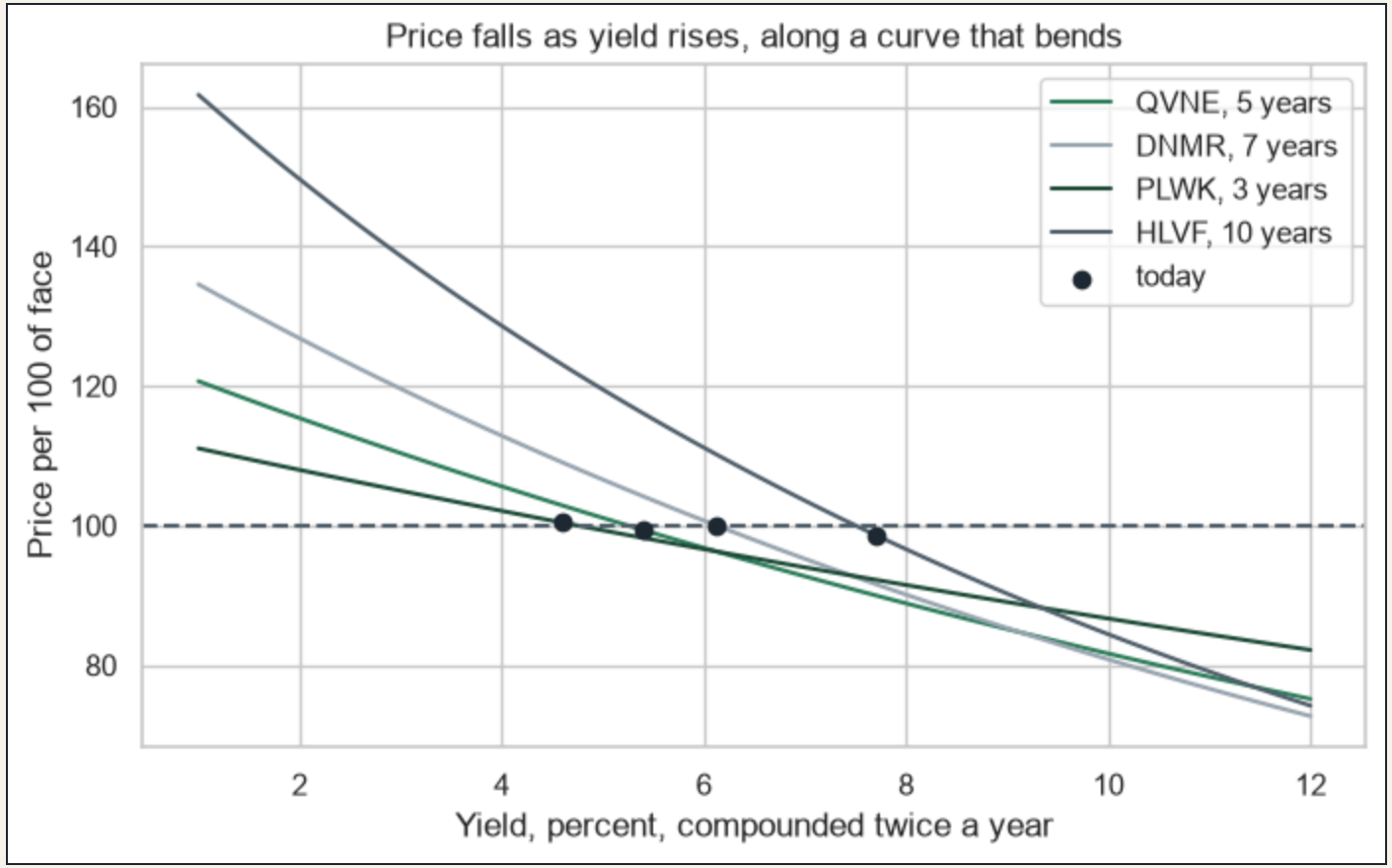
FIGURE 2 Pellwick's six extra coupons sum to 0.4159 over 100; Halvering's twenty shortfalls sum to minus 1.3773.

THE EXTRA COUPON, VALUED

	ticker	coupon_minus_yield	extra_pv
0	QVNE	-0.15	-0.649673
1	DNMR	0.00	0.000000
2	PLWK	0.15	0.415887
3	HLVF	-0.20	-1.377264

A premium is extra coupon, valued at the yield; a discount is the shortfall. The coupon decides the side of 100, the term how far.

3.2 The Price-Yield Curve, Charted



AGAINST EXCEL'S PRICE

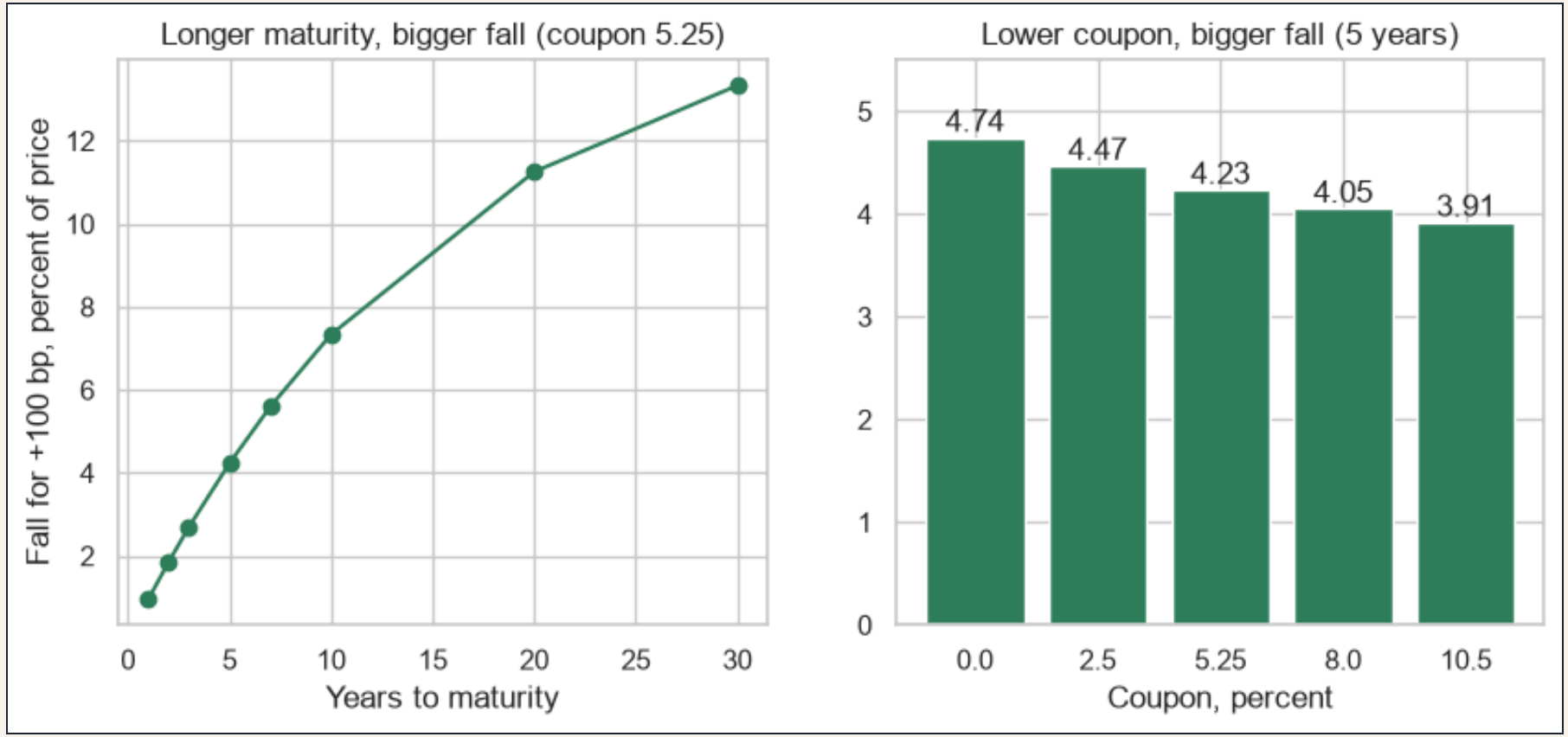
```
Seven PRICE rows, largest gap 5.7e-14 per 100 of face
```

EXCEL SIDE BY SIDE

JOB	EXCEL
The chart	select A1:B8, then Insert, Scatter with Straight Lines

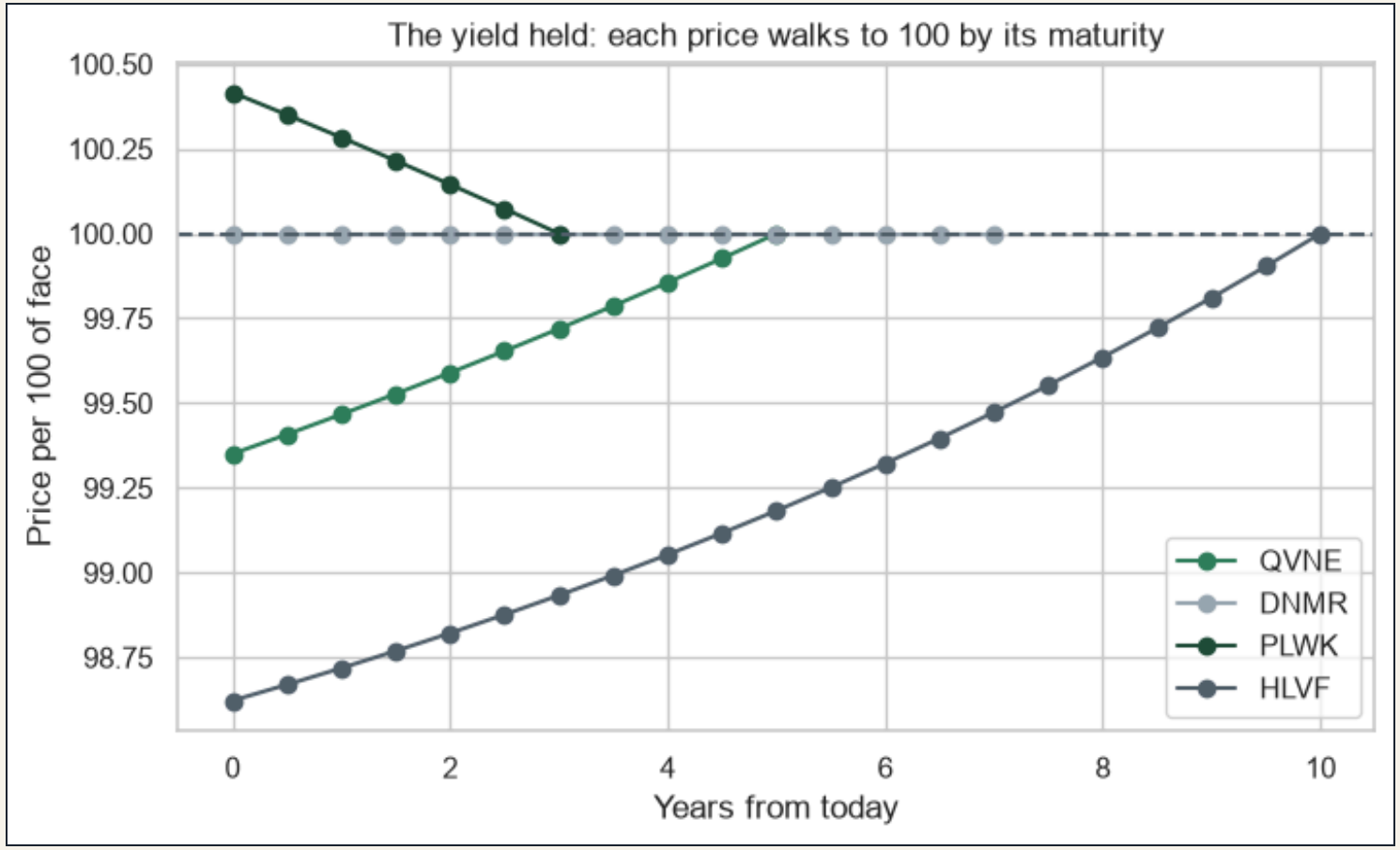
Your module and Excel's `PRICE` give the same column: the largest gap is in the fourteenth decimal.

3.3 Coupon and Maturity: Which Bond Moves More



Thirty years falls 13.33 percent, not thirty times the 1-year 0.95.

3.4 Pull to Par on Coupon Dates



HALVERING'S WALK

```
10 years left: 98.6227
 9.5 years left: 98.6697
   5 years left: 99.1828
 0.5 years left: 99.9037
```

EXCEL SIDE BY SIDE

JOB	RESULT
Halvering's walk	98.622736 to 99.903707
Pellwick's walk	100.415887 to 100.073314

Pellwick, the premium, walks down; Quorvane and Halvering walk up; Dunmarrow stays on the line.

3.5 One Yield, Three Quotes

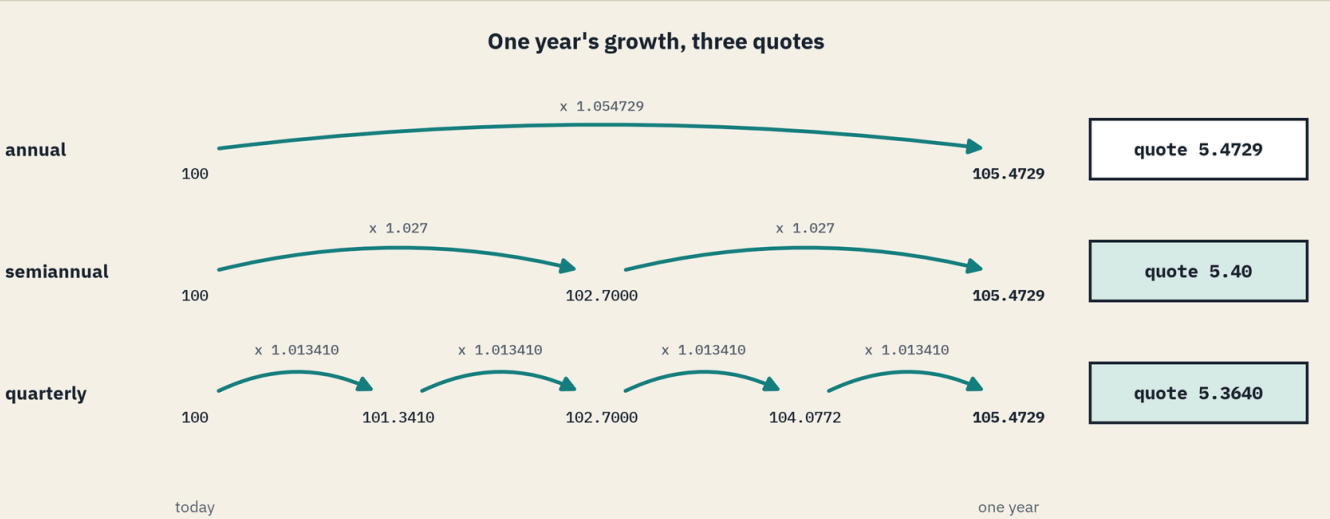


FIGURE 4 100 grows to 105.4729 in a year: annual 5.4729, semiannual 5.40, quarterly 5.3640.

ONE YEAR'S GROWTH

```
growth = 100 * (1 + 5.4 / 100 / 2) ** 2
annual_pct = growth - 100
```

FOUR TIMES A YEAR

```
Quarterly: 5.364034 percent
```

Compounding more often needs a lower quote for the same growth: 5.4729 annual, 5.40 semiannual, 5.3640 quarterly.

3.6 convert_yield in Your Module

THE FUNCTION

```
def convert_yield(yield_pct: float, from_frequency: int, to_frequency: int) -> float:
    for frequency in (from_frequency, to_frequency):
        if frequency not in (1, 2, 4):
            raise ValueError(f"frequency must be 1, 2, or 4, not {frequency}")
    growth_one_year = (1 + yield_pct / 100 / from_frequency) ** from_frequency
    period_rate = growth_one_year ** (1 / to_frequency) - 1
    return period_rate * to_frequency * 100
```

RESTATE THE FOUR

	ticker	annual_pct	yield_pct	quarterly_pct
0	QVNE	5.4729	5.400	5.3640
1	DNMR	6.2188	6.125	6.0788
2	PLWK	4.6529	4.600	4.5738
3	HLVF	7.8482	7.700	7.6273

EXCEL SIDE BY SIDE

Row **`tvm_nominal_03`** is **`=NOMINAL(6.25/100,2)`**, 0.061553; the test compares **`convert_yield(6.25, 1, 2)`** with 6.1553.

8 passed: `convert_yield` **agrees with Excel on all 22 rows.**

3.7 A Commit Message That Says Why

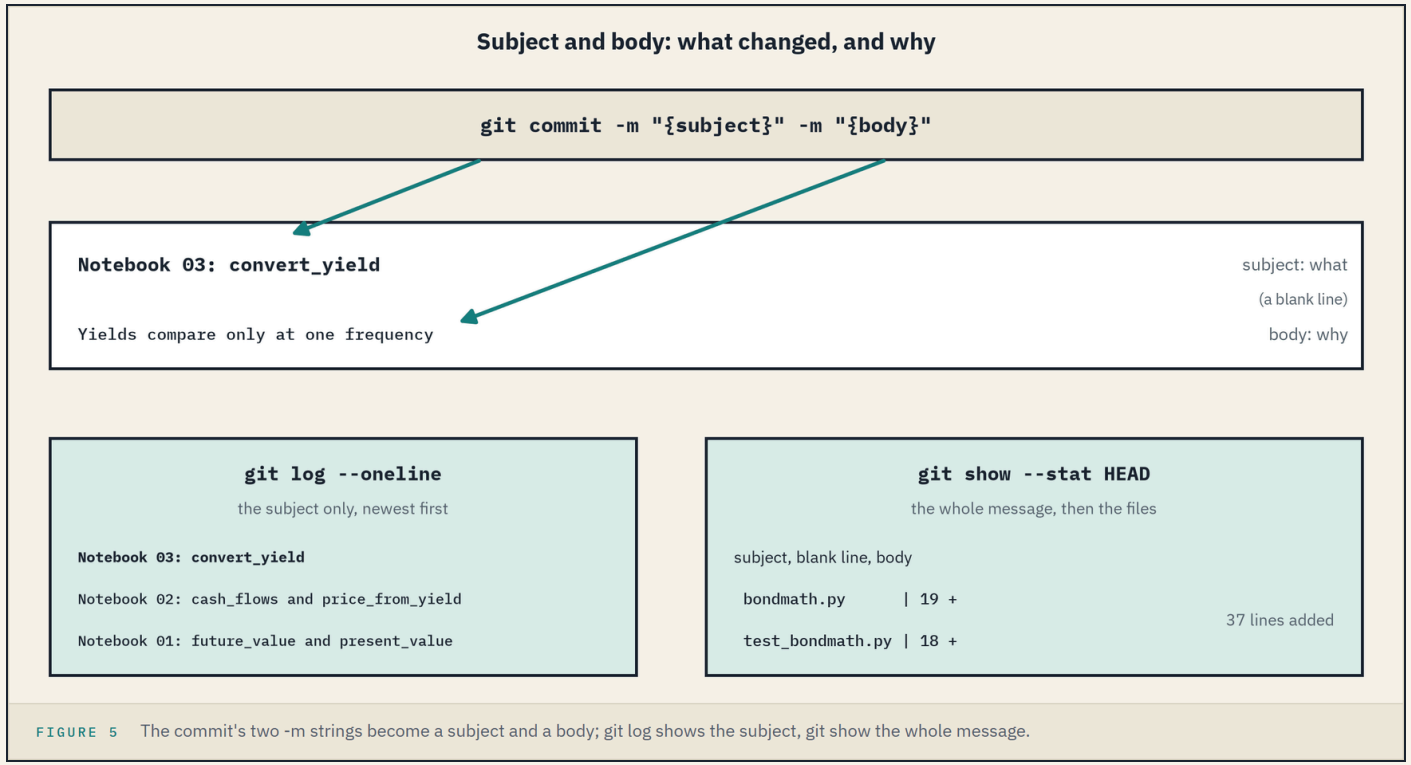


FIGURE 5 The commit's two -m strings become a subject and a body; git log shows the subject, git show the whole message.

SUBJECT AND BODY

```
subject = 'Notebook 03: convert_yield'
body = 'Yields compare only at one frequency'
!git -C "{work}" add bondmath.py test_bondmath.py
!git -C "{work}" commit -m "{subject}" -m "{body}"
!git -C "{work}" log --oneline
```

```
[main 4cdfc40] Notebook 03: convert_yield
 2 files changed, 37 insertions(+)
4cdfc40 Notebook 03: convert_yield
408a678 Notebook 02: cash_flows and price_from_yield
d029474 Notebook 01: future_value and present_value
```

Three commits, each subject saying what changed; the body keeps the why for whoever reads it next.

Bring It Home

IN POWERSHELL

```
# the work folder the setup cell printed
$work = "$env:TEMP\pfi_course2_03_k4w7q2xr"
Set-Location "$HOME\projects\my_bondmath"
Copy-Item "$work\bondmath.py", "$work\test_bondmath.py" .
python -m pytest -q
git diff --stat
git add bondmath.py test_bondmath.py
git commit -m "Notebook 03: convert_yield" -m "Yields compare only at one frequency"
git log --oneline
git show --stat HEAD
```

`my_bondmath` **holds five functions, eight tests, and three commits.**

You Can Now

- Say from the coupon and the yield whether a bond prices at a premium, at par, or at a discount, and value the premium as the coupon's excess.
- Draw a price-yield curve in Python and in Excel, and say why it slopes down and bends.
- Say which bond moves more for the same yield change: the longer one, and the one with the lower coupon.
- Show the pull to par, and that a bond earns its yield when its yield holds.
- Restate a yield at another frequency with **`convert_yield`**, and in Excel with **`EFFECT`** and **`NOMINAL`**.
- Write a commit message with a subject that says what and a body that says why.

NEXT
WEEK 1 · DAY 4

Yield From Price

A quoted price turned back into its yield by a search that halves a bracket, checked against Excel's Goal Seek, RATE and YIELD, tested for the prices it must refuse, and a bad edit undone with Git.